SARCache: Controlling Forecast Age in Diffusion Transformers Supplementary Material

Anonymous CVPR submission

A. Supplementary methods

A.1. Residual interface and state updates

The Qwen adapter projects packed image latents with `img_in`, extracts the target-image token suffix from the last transformer block, and stores its difference from the input embedding. Reuse applies `norm_out`/`proj_out` to the reconstructed representation with the current timestep embedding. Krea uses the corresponding image-token suffix and its current `final_layer`. The supported single-branch settings are `true_cfg_scale=1` for Qwen and `guidance_scale=0` for Krea. Each API defines its own numerical guidance convention.

The prediction history contains only residuals from full evaluations. A refresh replaces the oldest anchor, resets accumulated drift and clears the consecutive-reuse count. The adjacent-step indicator reference, by contrast, updates on every step. Keeping these histories separate prevents repeated comparison with an increasingly old probe. Algorithm S1 gives the ordering for the image sampler. The explicit forecast window supersedes the historical order codes 0/1/2, which correspond to $a = 0/\infty/2$. Code 2 does not mean quadratic prediction.

Algorithm S1 SARCache image sampling with explicit cache state

Require: x_0, schedule, condition c; gate, δ, w, q, forecast window a
Initialize empty residual history; $A \leftarrow 0, n \leftarrow 0$
for $i = 0, \ldots, N-1$ **do**
  Compute d_i ($d_0 = 0$); $A \leftarrow A + d_i$
  Update the adjacent-step indicator reference
  $b \leftarrow (i = 0) \vee (i = N-1) \vee (i < w)$
  $b \leftarrow b \vee (A > \delta) \vee (q > 0 \wedge n \geq q)$
  if b **then**
    $(v_i, e_i, r_i) \leftarrow \mathrm{Full}(x_i, \sigma_i, c)$
    Update residual history and full-pair statistics
    $\widetilde{v}_i \leftarrow v_i$; $A \leftarrow 0, n \leftarrow 0$
  else
    $\widehat{r}_i \leftarrow \mathrm{Predict}(\text{residual history}, i, a)$
    $\widetilde{v}_i \leftarrow H_\theta(E_\theta(x_i) + \widehat{r}_i, \sigma_i)$
    $n \leftarrow n + 1$
  end if
  $x_{i+1} \leftarrow x_i + (\sigma_{i+1} - \sigma_i)\widetilde{v}_i$
end for
return decoded x_N

A.2. Spectral and propagation-rate indicators

For spatial frequency f along one axis, the image filter uses

$$S(f) = \frac{1}{|f|^2 + \epsilon_f}, \tag{1}$$

$$G_i(f) = \frac{\alpha_i S(f)}{\alpha_i^2 S(f) + \beta_i^2 + \epsilon_f}, \qquad \alpha_i = 1 - \sigma_i, \quad \beta_i = \sigma_i, \tag{2}$$

Table S1. Sampling settings. Image runs use the single-branch guidance conventions described in the text. H3 retains its native audio–video solver.

Setting	Output	Steps	Guidance
Qwen-1024	1024^2	40	1
Qwen-2048	2048^2	40	1
Krea-1024	1024^2	52	0
H3 INT8	1344×768, 124 frames	20	1

where $\epsilon_f = 10^{-16}$. The two-dimensional gain is the product of the axis gains. It is normalized to unit mean when that mean is finite and positive. The indicator is the real inverse FFT of the gain multiplied by the latent FFT. At the initial zero-signal endpoint the gain can vanish. The mandatory first full evaluation avoids relying on a first difference there. Spatial filtering is computed in float32, with the result converted to the latent dtype and relative differences reduced at higher precision. The residual itself remains unfiltered.

An alternative gate weights embedding drift by an observed output-to-input change rate. For full evaluations at $k < j$,

$$\kappa_j = \frac{\langle |v_j - v_k| \rangle}{\max(\langle |e_j - e_k| \rangle, \epsilon)}, \qquad d_i^{\text{rate}} = \frac{\kappa_j \langle |e_i - e_{i-1}| \rangle}{\langle |v_j| \rangle + \epsilon},$$

with $\epsilon = 10^{-8}$ and mean absolute values denoted by angle brackets. The rate uses two full observations. The embedding difference uses adjacent steps. Before two full observations exist, the rate contribution is zero. This distinction requires separate full-step and adjacent-step histories.

A.3. H3 implementation and baselines

H3 retains the packed joint audio–video representation and the native output epilogue. Sea-H3 and SARCache-H3 reconstruct a hidden residual across the transformer stack. Their shared controller measures first-block modulated features, with a three-dimensional SEA filter over video time and space (stride four, exponent three) and an unfiltered audio probe. Video and audio drift accumulate separately. Their maximum controls a joint refresh. The main-test summaries show audio scores exceeding video scores in approximately 26% of SARCache-A gate observations, so audio is not treated as an inactive branch.

The H3 comparison arms differ in cached content. Tea-style uses the common SEA gate and holds the model output. It does not apply the original TeaCache probe and polynomial. Taylor uses Newton differences of full outputs in step-index coordinates. Easy-H3 uses raw embedding drift with a rate estimate and holds the output. Image EasyCache instead reconstructs a hidden residual and applies the current head. The H3 results refer to these project adaptations of the original methods.

No-reuse checks record exact latent agreement in 24 disabled-cache or wrapper comparisons across eight 56-frame pilot prompts, and in eight 124-frame wrapper comparisons. These checks support the unmodified path through the adapter. They do not measure the error of skipped evaluations. The native multistep solver, its history and audio conversion remain active when caching is enabled.

A.4. Residual geometry and propagation

The main-text propositions compare residual estimates at a common current state and common anchors. A full residual at this state can differ from the residual on the uncached trajectory. The analysis assumes Euclidean norms, deterministic model evaluations, and the prediction rule stated in the main paper. It does not assume that the latent indicator measures residual error.

Proof and interpretation of Proposition 1. Write $u = r_i^\star - r_j$ and $v = (r_j - r_k)/(j-k)$. Holding has error u, while a forecast of length b has error $u - bv$. Expanding the squared norm gives

$$\|u - bv\|_2^2 = \|u\|_2^2 - 2b\langle u, v\rangle + b^2\|v\|_2^2.$$

For $b > 0$, holding has no larger squared error exactly when $2\langle u, v\rangle \le b\|v\|_2^2$. After the window, Clipped uses $b = a$ and Unbounded uses $b = d > a$. Hence the condition with $b = a$ suffices for both comparisons. If $v = 0$, all three estimates agree. For $v \neq 0$, define the displacement along the historical trend as $c = \langle u, v\rangle / \|v\|_2^2$. The comparison reduces to $c \le b/2$. Components of u orthogonal to v cancel from the difference in squared errors.

For a continuing linear trend, $u = dv$, Unbounded is exact and returning to the observation incurs error $d\|v\|_2$. If the residual returns to its anchor, $u = 0$, holding is exact and extrapolation incurs error $b\|v\|_2$. These examples show why the preferred policy depends on residual dynamics. The criterion requires the current full residual and serves as an analytical diagnostic. SARCache selects its window before sampling.

Proof and scope of Corollary 1. For the uncertainty set $\{u : \|u\|_2 \le R\}$, the triangle inequality gives $\|u - bv\|_2 \le R + b\|v\|_2$. For nonzero v, equality holds at $u = -Rv/\|v\|_2$. If $v = 0$, the maximum is R. More generally, any estimate $r_j + w$ has worst-case error $R + \|w\|_2$ on this ball, minimized by $w = 0$. The uncertainty set is centered on the latest observation and contains no preferred direction. It is an explicit modeling assumption, not an estimated confidence region for these models. This result motivates fallback after directional information becomes unreliable. It neither selects a numerical horizon nor asserts that holding is best while a recent trend remains informative.

Proof of Proposition 2. Initially the states and caches agree. Suppose they agree before a step. At a prescribed refresh, deterministic full computation produces equal outputs and anchors. At a reuse step with one anchor, all policies hold. With two anchors and age $d \le a$, Finite and Unbounded use $r_j + dv$, and Clipped uses $r_j + \min(d, a)v = r_j + dv$. Equal outputs give equal next states. Induction proves equality of the complete trajectories. With a shared deterministic controller, equal states also give equal probes, controller histories and refresh decisions. A positive cap $q \le a$ ensures every reuse age remains within the window. The cap value zero disables this argument because it denotes no cap in the implementation. The result assumes that only the prediction policy changes, with the remaining settings and initial randomness held fixed.

Proposition 3 (propagation through the sampler). Let $F_i(x)$ be the full model output, $\widetilde{x}_i$ the approximate trajectory and x_i the uncached trajectory with the same initial state. Assume F_i is L_i-Lipschitz on a domain containing both states. At fixed $\widetilde{x}_i$, assume the current head is K_i-Lipschitz in its reconstructed hidden representation. Define $\epsilon_i = \|\widehat{r}_i - r_i^\star(\widetilde{x}_i)\|_2$ at a reuse step and $\epsilon_i = 0$ at a full evaluation. If $h_i = |\sigma_{i+1} - \sigma_i|$, Euler sampling satisfies

$$e_{i+1} \le (1 + h_i L_i) e_i + h_i K_i \epsilon_i, \qquad e_i = \|\widetilde{x}_i - x_i\|_2, \tag{3}$$

$$e_N \le \sum_{i=0}^{N-1} h_i K_i \epsilon_i \prod_{\ell=i+1}^{N-1} (1 + h_\ell L_\ell). \tag{4}$$

An empty product equals one.

Proof. Add and subtract $F_i(\widetilde{x}_i)$ in the difference of the two Euler updates. At the common state, the input embedding is identical and the head's input difference is the residual error, so $\|\widetilde{F}_i(\widetilde{x}_i) - F_i(\widetilde{x}_i)\|_2 \le K_i\epsilon_i$. Apply the triangle inequality and the Lipschitz bound for F_i to obtain the recurrence. Expanding it from $e_0 = 0$ gives the sum. A full evaluation sets the local forcing term to zero while retaining the accumulated state error.

The bound is conditional on finite Lipschitz constants over the visited domain. Their values are not estimated here. The head can amplify directions unequally. For example, the linear head $\mathrm{diag}(10, 1)$ maps residual errors $(0.2, 0)$ and $(0, 1)$ to output errors of norms 2 and 1, reversing their residual-norm ordering. Moreover, the available diagnostics use relative residual L1 and relative output L2 norms, whereas Proposition 1 uses absolute residual L2. The bound concerns latent error under Euler sampling. Converting it to a decoded-image or LPIPS guarantee would require additional assumptions on the decoder and perceptual mapping. H3 uses its native multistep solver and is outside this Euler bound.

A.5. Checks using retained data and numerical identities

The accompanying script `analysis/check_theory.py` verifies the algebra with seeded CPU calculations and reads the existing development CSV. It requires no model weights, inference or training. Synthetic vectors check the squared-error identity, the boundary attaining the worst-case bound, and the two counterexamples above. A scalar Euler example checks the propagation bound. These calculations check implementation of the algebra and complement the proofs.

The existing Qwen masks use $\{0, \ldots, 7\} \cup \mathrm{range}(8, 40, g+1) \cup \{39\}$. Their maximum reuse ages are g. Table S2 checks the two measured equivalences implied by Proposition 2. Equality is checked on the exported aggregate values, whose precision limits remain applicable. The additional $g2$ windows 3 and 4 in the main figure are analytic aliases.

Table S2. Existing development data used to check policy equivalence. Differences are finite window minus Unbounded, evaluated from the retained CSV. Each arm has 160 samples.

Schedule	Maximum reuse age	Window	Δ LPIPS	Δ SSIM	Δ PSNR
$g2$	2	2	0	0	0
$g4$	4	4	0	0	0

The retained diagnostic aggregates omit the per-step Euclidean inner products required by Proposition 1. They therefore cannot establish how often its condition holds. Evaluating that frequency would require suitable saved tensors or new inference. The present revision adds proofs and checks of existing records without claiming such a measurement.

B. Original frozen experiment: prompts and configurations

B.1. Partitions and coverage

The image pool has 348 prompts: 186 from PartiPrompts, 43 from GenEval and 119 authored stress prompts. It is partitioned into eight pilot, 60 calibration, 80 Qwen-development and 200 main-test prompts. Krea and Qwen-2048 use prespecified 100-prompt test subsets. The video pool has 408 prompts: 99 from VBench-1 sources, 180 from VBench-2 sources and 129 authored prompts. Its partitions contain eight pilot, 60 calibration, 40 development, 200 main-test, 60 long-test and 40 precision-test prompts. The latter two sets have no completed results in the present analysis. The VBench source revision is `fd18b3d055cb0fc6f066ca90fe2c3c8cbb698490`.

Deduplication normalizes case and punctuation, replaces digits, and checks exact matches, content words and word-ngram overlap against earlier prompts and within the pools. Scenario families combine a category with a content-word multiset hash. Identifiers, text hashes and recorded families are disjoint across the partitions. These checks remove the specified forms of overlap but cannot establish semantic independence of every paraphrase or template.

Image main tests use seeds 0, 1 and 2. H3 uses seed 0. Development tests use seeds 0 and 1, calibration uses seed 10, and pilot tests use seed 20. The final records contain 9,600 image candidate metric rows and 2,600 H3 jobs, including 200 uncached references. Each image arm has its expected 600 or 300 unique prompt–seed pairs, and every H3 arm has 200. All candidate quality values are finite. The reference-pairing record reports 1,200 matching full-reference PNG pairs across the two image runners. Final coverage includes resolved execution retries and should not be read as an absence of failed submissions.

B.2. Calibration and confirmation

Calibration separates an initial screen, 24-prompt selection and 60-prompt confirmation. Confirmation does not reopen selection, and test results do not retune the chosen parameters. Bands A and B initially target $2\times$ and $2.5\times$ acceleration with a $\pm 5\%$ timing tolerance. A selected configuration can therefore remain outside its target band at confirmation or test.

Four entries in the main image table were selected outside their calibration bands: EasyCache at Krea-1024 A, TeaCache and EasyCache at Qwen-2048 B, and TeaCache at Krea-1024 B. They are retained with their measured test times. A/B denote the intended calibration groups.

For Qwen-1024 A, the target was relaxed from 3.1532 to 3.3529 s for common reachability. SARCache confirmed at 3.569 s, 6.4% above that target, and tested at 3.529 s. SeaCache confirmed at 3.300 s and tested at 3.292 s. This difference in latency is retained in the main comparison.

H3 targets are 226.8 and 181.4 s. Easy-H3 took 257.5 s at selection and 238.9 s at confirmation, above the A upper bound of 238.14 s. It thus retains its calibration-unmatched label. At test, however, its 228.4 s mean is inside the 215.46–238.14 s interval. Both its test fidelity and this change in band status matter to the comparison. The reduced-step arm takes 239.3 s, just outside that interval.

B.3. Forecast windows and reuse caps

All six frozen image SARCache configurations use the spatial image-latent SEA indicator and raw residual reconstruction. The table lists the threshold δ, warmup w, cap q and window a. "Permitted" indicates that the cap allows a fallback to holding. Activation frequency requires step traces.

Setting	Band	δ	Warmup w	Cap q	Window a	Fallback active?
qwen1024	A	0.12	8	2	3	Impossible ($q \leq a$)
qwen1024	B	0.4	8	4	4	Impossible ($q \leq a$)
qwen2048	A	0.18	2	8	2	Permitted
qwen2048	B	0.27	8	6	1	Permitted
krea1024	A	0.3	16	4	4	Impossible ($q \leq a$)
krea1024	B	0.3	10	4	4	Impossible ($q \leq a$)
H3	A	0.3162	6	4	3	18.18%
H3	B	0.3162	4	8	1	84.62%

Qwen-1024 and Krea cannot activate the finite-window fallback because $q \leq a$. Qwen-2048 permits it, but its final-test step traces are unavailable for recounting the frequency. For H3, the recorded fractions of eligible reuses falling back to holding are 0.1818 and 0.8462 at A and B. A holding-only policy has fraction one by this definition, without performing finite-window prediction.

C. Metrics and statistical analysis

C.1. Image, temporal and audio metrics

Image measurements use RGB values in $[0, 1]$. LPIPS uses AlexNet v0.1 with inputs mapped to $[-1, 1]$. Model-weight hashes are fixed by the evaluator. SSIM uses Gaussian weights with $\sigma = 1.5$, channel averaging, population covariance and `data_range=1`. PSNR is $10 \log_{10}(1/\mathrm{MSE})$ over all RGB pixels, averaged after calculating the score for each sample. Alpha channels are excluded.

For H3, 12 frame indices are obtained by rounding an evenly spaced grid across each clip. LPIPS and SSIM are averaged over those frames. PSNR uses pooled RGB pixel MSE across the selected frames. Temporal-reference error uses all MP4-decoded frames:

$$E_{\mathrm{temp}} = \mathrm{mean} \left| (I_{t+1} - I_t) - (I^{\mathrm{ref}}_{t+1} - I^{\mathrm{ref}}_t) \right| .$$

Pixel values are in $[0, 1]$. This measures deviation from the reference's frame-to-frame changes, not intrinsic motion quality. Audio distance is the mean absolute difference of log-STFT magnitudes from paired WAVs, using a Hann window, FFT size 1024, hop 256, no centering and a 10^{-5} log offset. It does not establish audio preference or synchronization quality.

C.2. Video source asymmetry

All 200 references use retained uint8 frames. Of 2,400 candidates, 342 use recovered VAE-decoded uint8 frames and 2,058 use H.264 MP4 frames (YUV420p, CRF 18). SARCache-A has 26 frame-array and 174 MP4-source candidates. Sea-A has 48 and 152. Converting both sources to uint8 does not remove compression differences.

For the 2,058 MP4-source cross-checks, post-hoc minus in-job LPIPS averages +0.002956, with mean absolute change 0.003689 and range $[-0.008435, +0.022859]$. The mean change is +0.005320 for SARCache-A and +0.003361 for Sea-A. These method-dependent shifts characterize pipeline sensitivity. They are neither a universal offset nor an error bound. Temporal and audio cross-checks use the same MP4/WAV inputs and do not provide independent lossless validation.

C.3. Paired inference

Measurements are paired by prompt and seed. We use 10,000 prompt-cluster bootstrap draws, retaining all seeds of a sampled prompt, and percentile 95% intervals. The two-sided test uses null-centered differences. RNG seeds are derived deterministically from comparison labels, with master seeds 20261010 for images and 20261014 for H3. All 153 comparison–metric rows reproduce the tabulated means, differences, intervals and probabilities to the reported six-decimal precision.

The image Holm family contains six SARCache–Sea comparisons across three metrics. The H3 family contains eight pairs across three metrics. Temporal and audio tests remain exploratory. This implemented analysis differs from the initial plan of two primary comparisons and 2,000 draws, and is reported without an independently verified preregistration claim. The smallest adjusted H3 probabilities are 0.0024.

Setting	Band	SAR LPIPS	Sea LPIPS	Δ	95% CI	Adjusted p
qwen1024	A	0.0115	0.0225	-0.0111	[-0.0119, -0.0103]	0.0018
qwen1024	B	0.0496	0.0435	0.0061	[0.0040, 0.0082]	0.0018
qwen2048	A	0.0261	0.0247	0.0014	[0.0005, 0.0023]	0.0112
qwen2048	B	0.0693	0.0665	0.0027	[-0.0002, 0.0056]	0.2592
krea1024	A	0.0492	0.0486	0.0007	[-0.0005, 0.0020]	0.8774
krea1024	B	0.0570	0.0455	0.0115	[0.0068, 0.0163]	0.0018
H3	A	0.1866	0.3721	-0.1855	[-0.1982, -0.1733]	0.0024
H3	B	0.3138	0.3106	0.0033	[0.0029, 0.0037]	0.0024

Differences are SARCache minus the corresponding Sea implementation. These intervals describe prompt sampling under the measured pipeline. They exclude calibration selection uncertainty and media-source bias. The best development windows are exploratory choices, not independently confirmed optima. Image timing intervals cannot be reconstructed from aggregate run times. H3 source data include both ratios of mean times and mean paired speedups with prompt intervals.

D. Fixed-refresh and configuration experiments

D.1. Qwen controls

The zero-based refresh positions are

$$\mathcal{M}_{g4} = \{0, \ldots, 8, 13, 18, 23, 28, 33, 38, 39\}, \qquad \mathcal{M}_{g8} = \{0, \ldots, 8, 17, 26, 35, 39\}.$$

They give 16 and 13 full evaluations. Each predictor uses its own trajectory and full-evaluation history. Refresh-position checks reconstructed from full-call tags report the prescribed mask for all 1,920 outputs. Under $g4$, the maximum reuse age is four: the 160 metric triplets for window 4 and unbounded prediction coincide, and the pairing record reports identical PNGs. The lowest means occur at windows 3 and 4 for $g4$ and $g8$, respectively. Because selection and comparison use the same development set, those minima remain exploratory.

D.2. Recalibrated image variants

The Qwen baseline uses $(\delta, w, q, a) = (0.12, 8, 2, 3)$. Holding and unbounded prediction use threshold 0.15 after recalibration. The one-step-warmup variant also uses 0.15. Its metrics match the unbounded variant, with 160 identical PNG pairs reported. Since the baseline satisfies $q < a$, removing its finite window alone would not change its predictions. The observed difference after recalibration must involve other changed settings.

The raw-embedding arm disables the image-latent probe while leaving embedding filtering off. It changes both probe domain and filtering. The no-cap arm uses threshold 0.11 and confirms 8.2% above its calibration target. Each variant has 160 development measurements. These coupled changes and timing differences prevent attribution of a separate benefit to each component.

Table S3. Recalibrated Qwen-1024 variants. Development set, 80 prompts × two seeds. The changed gate uses raw embeddings, so both filtering and probe domain change.

Variant	Time (s)↓	LPIPS↓
Frozen SARCache-A	3.503	0.0109
Hold	3.273	0.0232
Unbounded linear	3.273	0.0137
Raw embedding gate	3.422	0.0167
No reuse cap	3.578	0.0122
One-step warmup	3.272	0.0137

Thresholds were recalibrated. These are not identical-latency or fixed-schedule comparisons. The configuration without a reuse cap missed its calibration budget. Linear and one-step-warmup outputs were reported pixel-identical.

E. Runtime and reproducibility

H3 request time adds the recorded pre-decode total to both decoder durations, averaged over 200 jobs per arm. Initial per-process checkpoint loading and media writing are excluded. Transfers and model offloading within the native request remain

part of its cost. These original H3 summaries do not provide isolated gate cost, peak-memory comparisons, repeat-only latency stability or per-step refresh positions. The new Qwen cost summaries have a separate scope and are described below.

The original records retain 9,600 image test measurements, 2,600 H3 jobs, 2,880 fixed-refresh measurements and 960 variant measurements, with partition manifests and implementation snapshots. These are separate from the October 18 additions. Available statistics are recomputed without new inference or pixel-level metric recomputation. Clean multiseed video, long-video and precision transfer, human preference and standard VBench scores remain incomplete.

F. Additional controlled experiments and evidence audit

The October 18 additions comprise Qwen controls (E1), operating points (E2), case indices (E3), costs (E4), Krea transfer (E5), and repaired H3 development metrics. E6 main results remain incomplete. No new inference was run during manuscript preparation.

F.1. Integrity, precision and confirmation exposure

All 18 files listed in the supplied freeze manifest match their byte-level SHA-256 hashes. Partition-manifest hashes instead use normalized text: newline normalization explains their difference from the byte hashes. The confirmation partition contains 80 identifiers, with eight categories of ten prompts and seeds 0/1. Its manifest discloses two older development exposures, `cnf_0032` and `cnf_0033`. These are retained in the main analysis and removed only in a separately labelled sensitivity analysis.

Confirmation tests are reconstructed from all 160 paired rows per mask in the E3 candidate index, with LPIPS rounded to five decimals. The supplied statistics file contains only development results. We use the delivered bootstrap procedure: 10,000 prompt-cluster draws, deterministic comparison-label seeds, percentile intervals, null-centered two-sided tests and a six-comparison Holm family. Rounding alone can alter an arm mean by at most 5×10^{-6} and a mean paired difference by at most 10^{-5}. It can also change tail probabilities. Our $g4$ Finite–Clipped adjusted probability is 0.0031, whereas the campaign log reports 0.0037. The original full-precision input is unavailable. Tables explicitly report the reconstruction, not an asserted bitwise reproduction of the original test.

Table S4. All 80 confirmation prompts. Five-decimal-export reconstruction. Each prompt contributes two seeds.

Mask	Comparator	Finite mean	Comparator mean	Δ [95% CI]	p_H
g4	Hold	0.04334	0.05328	-0.0099 [-0.0116, -0.0085]	0.0006
g4	Unbounded	0.04334	0.04603	-0.0027 [-0.0037, -0.0017]	0.0006
g4	Clipped	0.04334	0.04455	-0.0012 [-0.0020, -0.0004]	0.0031
g8	Hold	0.07424	0.08892	-0.0147 [-0.0161, -0.0134]	0.0006
g8	Unbounded	0.07424	0.10492	-0.0307 [-0.0334, -0.0279]	0.0006
g8	Clipped	0.07424	0.08742	-0.0132 [-0.0149, -0.0114]	0.0006

Table S5. Sensitivity analysis: 78 prompts. Five-decimal-export reconstruction. Each prompt contributes two seeds.

Mask	Comparator	Finite mean	Comparator mean	Δ [95% CI]	p_H
g4	Hold	0.04335	0.05336	-0.0100 [-0.0117, -0.0085]	0.0006
g4	Unbounded	0.04335	0.04588	-0.0025 [-0.0036, -0.0015]	0.0006
g4	Clipped	0.04335	0.04443	-0.0011 [-0.0019, -0.0003]	0.0092
g8	Hold	0.07433	0.08901	-0.0147 [-0.0161, -0.0133]	0.0006
g8	Unbounded	0.07433	0.10490	-0.0306 [-0.0334, -0.0278]	0.0006
g8	Clipped	0.07433	0.08750	-0.0132 [-0.0150, -0.0113]	0.0006

Table S6. Per-pair boundaries. Counts where Finite has higher LPIPS. Denominator 160 per mask. No inference treats these pairs as independent prompts.

Mask	vs Hold	vs Unbounded	vs Clipped
g4	12	49	64
g8	0	4	12

The 78-prompt result is a post-hoc sensitivity analysis, not a replacement primary population. No samples were removed because of their performance. The selection rules are documented in the delivered manifests. An independent preregistration timestamp was not verified. The case index's "improve/tie/degrade" labels are quantile bins, not signs of improvement, equivalence tests or failure labels. We use actual signed differences when reporting per-sample outcomes.

F.2. Refresh masks and predictor controls

Qwen masks are $\{0, \ldots, 7\} \cup \mathrm{range}(8, 40, g+1) \cup \{39\}$ for $g \in \{2, 4, 6, 8, 10\}$. They contain 20, 16, 14, 13 and 12 full evaluations. The fixed masks bypass online gate, cap and warmup decisions. Only full evaluations update the two anchors. All four predictors share the current input projection and output head. Each evolves its own state. With only one anchor, each holds. The Clipped comparator is

$$\widehat{r}_i^{\mathrm{clip}} = r_j + \frac{\min(i-j, a)}{j-k}(r_j - r_k).$$

At $g2$, windows 2, 3, 4 and infinity are analytically equivalent. The delivered table contains measured $a = 2$ and Unbounded rows, while the other two display cells are aliases. At $g4$, $a = 4$ and Unbounded also coincide. These are distinct from the non-equivalent $g4$ confirmation policies Finite($a = 3$) and Clipped($a = 3$).

The freeze notes report image-reference parity, 128/128 probe-isolation checks, and fallback fractions of 25% and 44%. Missing PNGs and full job logs prevent independent verification beyond configuration hashes and statistics. The new system cap $q = 6 > a = 3$ permits fallback but does not measure its activation rate.

F.3. Shadow diagnostics by phase and carrier

The diagnostic CSV contains 64 aggregate cells representing 3,264 eligible reuse steps. Counts are retained in aggregation. The relative residual L1 error divides the absolute-error sum by the true-residual absolute sum. The output L2 error divides by the true-output norm. The four counterfactual estimates share state and anchor history within a carrier. Hold and Unbounded carriers are separate approximate trajectories. The supplied code uses a state-isolated true forward and records diagnostic time separately. Diagnostic runs are excluded from speed reporting. Aggregates cannot support an independently reconstructed prompt-cluster interval because their prompt identities have already been collapsed.

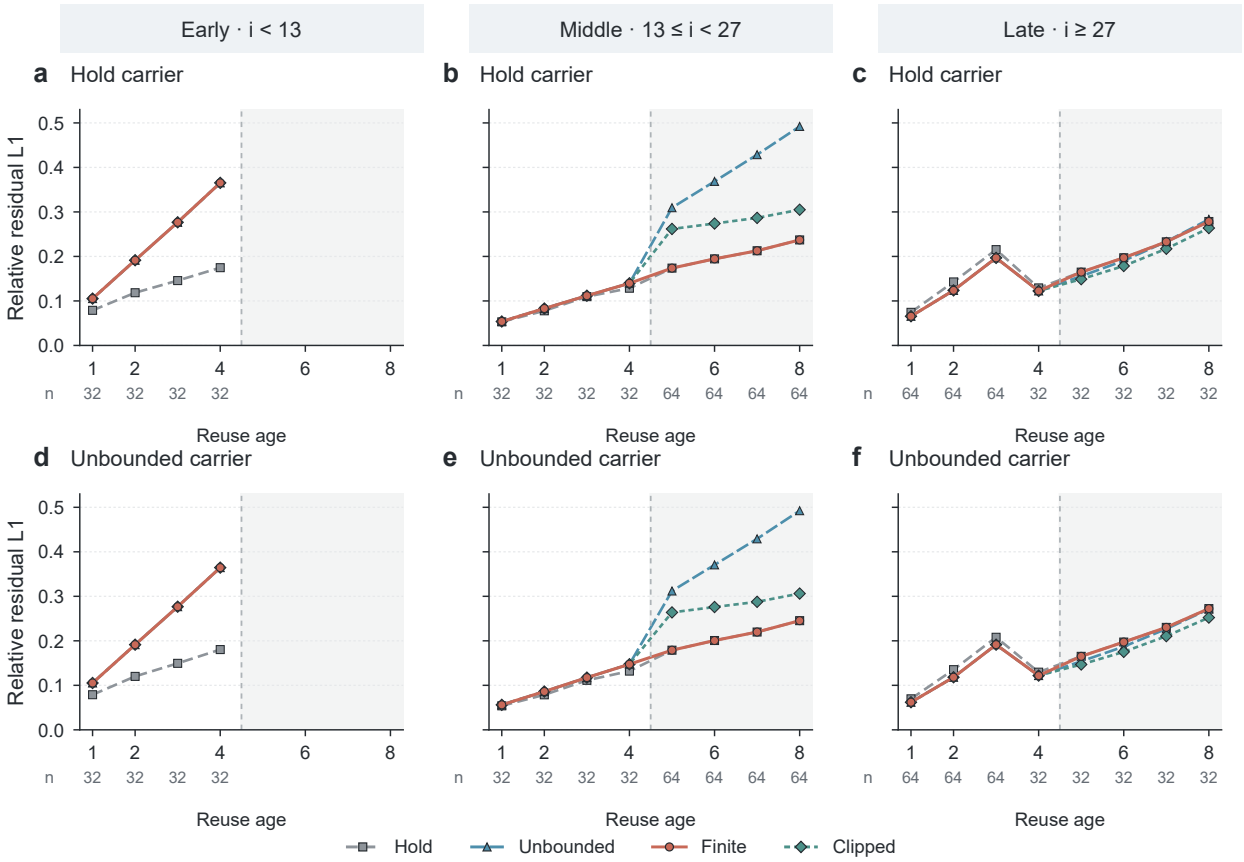


Figure S1. Phase- and carrier-specific residual errors at $g8$. Each point is an aggregate at the displayed age. Counts below each axis are aligned to observed ages. The implementation defines phases by step index: early $i < 13$, middle $13 \leq i < 27$, late $i \geq 27$. Missing phase–age combinations are not extrapolated. Anchor spacing and noise level also vary. These plots do not isolate a causal age effect.

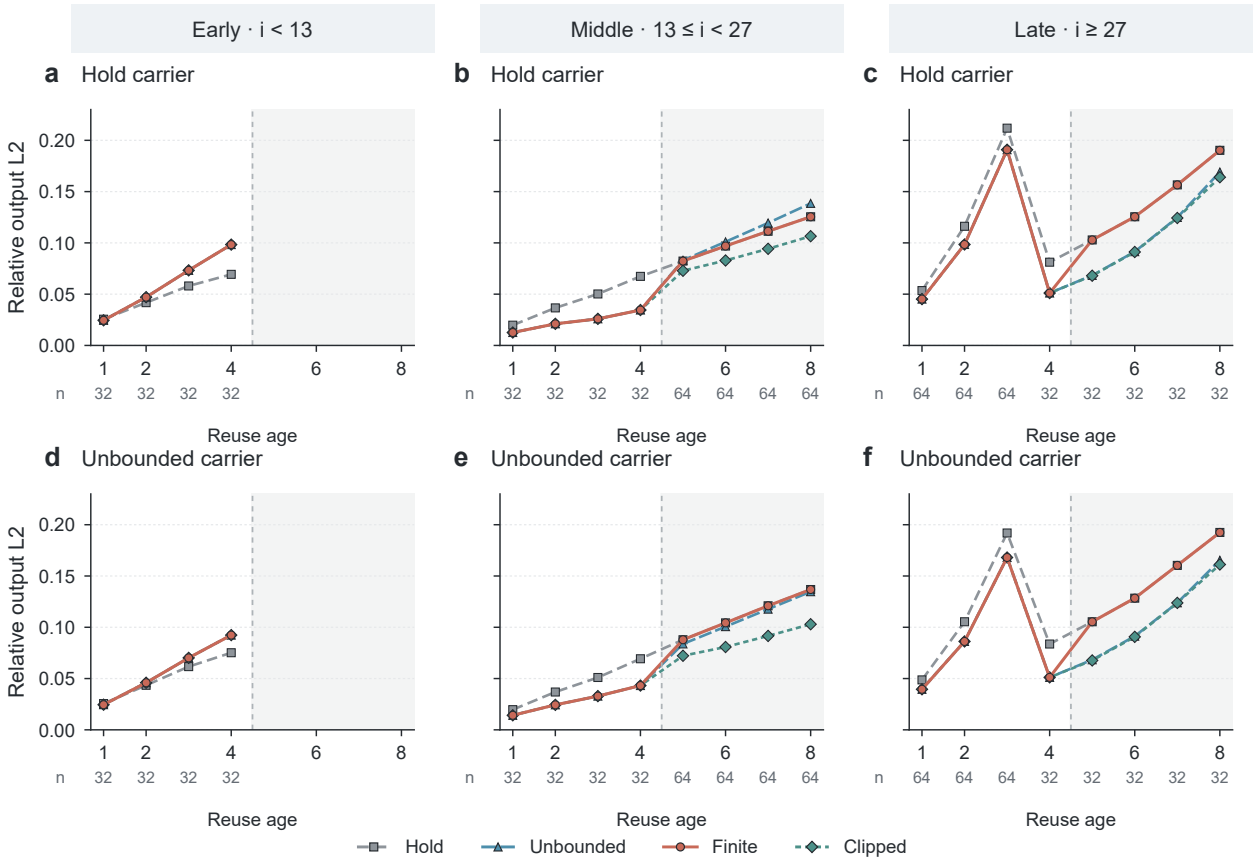


Figure S2. Output-space errors for the same diagnostic cells. The denominator and geometry differ from residual L1. A lower residual error does not ensure a lower current-head output error, and neither alone establishes lower final rollout LPIPS.

F.4. Complete measured operating points

The executed E2 manifest selects thresholds on calibration by absolute time-ratio distance. It permits a point within ± 0.05 of its target after the 24-prompt stage. This differs from the execution plan's relative $\pm 5\%$ tolerance. Both conventions and calibration/test status are retained in the source data. Table S7 shows all 25 measured configurations using actual time. Taylor uses second-order prediction of the output and varies the scheduled number of full calls.

Table S7. All E2 operating points. Speedup is mean reference time divided by mean method time. "Cal." is the supplied calibration status (F=frozen, O=off-band). "Test" indicates whether $|T/T_{\text{ref}} - t_{\text{target}}| > 0.05$ (Y=yes). Statuses are not interchangeable. LPIPS intervals and PSNR means are supplied aggregates. PSNR $+\infty$ is retained.

Arm	Time (s)	Speedup	Target	Actual	LPIPS [95% CI]	PSNR	Cal.	Test
sar_t40	2.329	2.697	0.40	0.371	0.07642 [0.07391, 0.07889]	30.352	F	–
sar_t45	2.636	2.383	0.45	0.420	0.04012 [0.03877, 0.04147]	33.457	F	–
sar_t55	3.399	1.848	0.55	0.541	0.01257 [0.01210, 0.01304]	38.569	F	–
sar_t65	4.005	1.569	0.65	0.637	0.00570 [0.00548, 0.00592]	41.732	F	–
sar_t75	4.617	1.361	0.75	0.735	0.00317 [0.00305, 0.00329]	44.079	F	–
tay_t40	2.608	2.409	0.40	0.415	0.19807 [0.18428, 0.21161]	21.270	F	–
tay_t45	2.912	2.157	0.45	0.464	0.20299 [0.19012, 0.21619]	20.821	F	–
tay_t55	3.527	1.781	0.55	0.561	0.12679 [0.11626, 0.13726]	24.430	F	–
tay_t65	4.139	1.518	0.65	0.659	0.15319 [0.14292, 0.16379]	22.306	F	–
tay_t75	4.749	1.323	0.75	0.756	0.11006 [0.10273, 0.11725]	24.104	F	–
ec_t40	2.489	2.518	0.40	0.397	0.05062 [0.04787, 0.05363]	30.909	F	–
ec_t45	2.897	2.163	0.45	0.462	0.03314 [0.03116, 0.03531]	32.738	F	–

Arm	Time (s)	Speedup	Target	Actual	LPIPS [95% CI]	PSNR	Cal.	Test
ec_t55	3.405	1.841	0.55	0.543	0.02041 [0.01900, 0.02193]	34.856	F	–
ec_t65	4.076	1.538	0.65	0.650	0.01058 [0.00978, 0.01150]	37.904	F	–
ec_t75	4.872	1.287	0.75	0.777	0.00471 [0.00424, 0.00524]	$+\infty$	F	–
sea_t40	1.920	3.265	0.40	0.306	0.09786 [0.09458, 0.10112]	26.919	O	Y
sea_t45	2.216	2.828	0.45	0.354	0.06819 [0.06508, 0.07131]	28.711	O	Y
sea_t55	2.785	2.251	0.55	0.444	0.03270 [0.03103, 0.03439]	33.204	O	Y
sea_t65	3.274	1.914	0.65	0.522	0.01387 [0.01323, 0.01453]	37.532	O	Y
sea_t75	4.170	1.503	0.75	0.665	0.00452 [0.00430, 0.00473]	42.243	O	Y
tea_t40	2.172	2.885	0.40	0.347	0.33497 [0.32070, 0.34932]	17.635	F	Y
tea_t45	2.569	2.439	0.45	0.410	0.29106 [0.27703, 0.30498]	18.470	F	–
tea_t55	3.050	2.055	0.55	0.487	0.21718 [0.20419, 0.23007]	20.514	O	Y
tea_t65	3.874	1.618	0.65	0.618	0.21375 [0.20052, 0.22713]	20.593	F	–
tea_t75	4.823	1.299	0.75	0.770	0.17156 [0.15819, 0.18519]	22.239	F	–

Each row reports 600 runs and 600 metric pairs. The delivered per-configuration means and LPIPS intervals are retained. The full run records and 25 per-image metric tables are not supplied. For the three configurations represented by the complete 600-row E3 system index, rounded metrics allow a selected-point exploratory reanalysis. SARCache `sar_t55` minus SeaCache `sea_t65` is -0.00130 (95% CI $[-0.00183, -0.00078]$), and minus EasyCache `ec_t55` is -0.00783 ($[-0.00922, -0.00663]$). These points have different mean latencies and were selected after the broader curve was available. They are not new prespecified equal-budget tests. No curve-level dominance or interpolated equal-latency improvement is inferred. All 25 CLIP fields are missing, and the referenced CLIP-score directory is absent, so the campaign notes' semantic-alignment claim is not adopted.

F.5. Krea transfer and resource-cost boundaries

The Krea transfer uses 52 native steps, two seeds, 40 prompts and $a = 3$ without target-model reselection. Its masks are the corresponding $\{0, \ldots, 7\} \cup \mathrm{range}(8, 52, g+1) \cup \{51\}$, giving 18 and 14 full calls. The partition manifest reports five fallback admissions. No strict untouched-split claim is made. Only the rounded eight means in the campaign status are available, so this table is descriptive and has no reconstructed intervals.

Table S8. Krea transfer, mean LPIPS. Smaller is closer to the reference. Finite and Clipped both use $a = 3$.

Mask / full calls	Hold	Finite	Clipped	Unbounded
$g4$ / 18	0.2106	0.1000	**0.0880**	0.0902
$g8$ / 14	0.4013	0.3454	0.2258	**0.1736**

The cost experiment aggregates 16 calibration prompts, two seeds and five repeats, hence 160 timing observations per arm and protocol. There are separate Full controls in the two runners. The CSV standard deviations pool prompt variation and repeat variation. They are not per-prompt repeatability estimates or confidence intervals. The instrumented operations counter includes work beyond the horizon decision, including current-state reuse operations. Its average duration per recorded operation is approximately 0.24 ms for SARCache A/B, but no replay-mask gate-off ablation is provided. The label "ops per reuse" in the original report divides by recorded operations calls and should not be read as an isolated predictor timing.

The runner measures `torch.cuda.max_memory_allocated()/2**30`, so the reported unit is GiB. Full and SARCache A/B all round to 36.86 GiB. Equal rounded peaks do not prove zero cache memory. Reserved-memory measurements, baseline memory peaks and an isolated measured cache increment are absent. The approximately 235 MB cache estimate in campaign prose is not presented as a measured result. The older H3 system timings below do not measure the same costs.

F.6. Repaired H3 development experiment

The retrospective audit attributes the earlier $g4$ discrepancy to incompatible generation waves. It replaces 223 stale cells and retains 737 fresh cells. The new CSV contains 960 unique strategy–mask–prompt–seed rows: six policies, two masks, 40 prompts and two seeds. For $g4$, all 80 LPIPS differences between $a = 4$ and Unbounded are exactly zero, stronger than the

report's median-only check. This verifies metric equality, not retained-media identity. The audit records eight executed full calls for both masks. The campaign summary's seven-call $g8$ heading is discarded.

The recorded H3 masks are $\{0, 1, 2, 3, 4, 9, 14, 19\}$ and $\{0, 1, 2, 3, 4, 8, 13, 19\}$. Their maximum reuse ages are four and five. The legacy H3 names $g4/g8$ therefore do not share the literal maximum-age interpretation of the Qwen masks, and cross-model gap labels must not be equated.

Table S9. Repaired H3 development LPIPS. Both masks executed eight full calls, with 40 prompts and two seeds per cell.

Policy	$g4$ mean [95% CI]	$g8$ mean [95% CI]
Hold	0.2043 [0.1861, 0.2234]	0.2026 [0.1844, 0.2218]
Window 1	0.2061 [0.1872, 0.2265]	0.2046 [0.1866, 0.2235]
Window 2	0.2130 [0.1936, 0.2331]	0.2140 [0.1953, 0.2337]
Window 3	0.2339 [0.2129, 0.2560]	0.2696 [0.2475, 0.2934]
Window 4	0.3344 [0.3103, 0.3589]	0.2806 [0.2578, 0.3047]
Unbounded	0.3344 [0.3101, 0.3596]	0.2969 [0.2743, 0.3197]

Means and 95% prompt-cluster intervals are recomputed from the 960-row table, retaining seeds 0/1 together. Hold has the lowest mean under both masks. This does not establish its optimality under other video schedules.

The old mixed-generation diagnostic table is superseded, not averaged with the replacement. The original H3 system table below is a separate 200-prompt, seed-zero experiment with mixed candidate media. E6 job manifests describe a pilot and additional seed-one work, but no completed main metric table is supplied. Pilot prose cannot establish multiseed main performance, and a unified post-hoc decoder does not itself guarantee lossless source equivalence. We do not relabel the old system table as clean multiseed evidence.

Table S10. H3 INT8 deployment at frozen operating points. Each row uses 200 prompts and seed zero, at 1344×768, 124 frames, 24 fps. Cache arms retain 20 solver steps. Metrics use the mixed-source video evaluation described in the supplementary evaluation protocol.

Implementation	Band	Time (s)↓	Speed↑	LPIPS↓	SSIM↑	PSNR↑	E_{temp} ↓
Uncached	ref.	441.1	1.00×	ref.	ref.	ref.	ref.
SARCache-H3	A	222.9	1.98×	<u>0.1866</u>	<u>0.7569</u>	<u>22.37</u>	<u>0.0277</u>
Sea-H3		230.2	1.92×	0.3721	0.5685	16.59	0.0400
Tea-style output hold		230.5	1.91×	0.3733	0.5679	16.56	0.0400
Taylor output predictor		235.1	1.88×	0.5151	0.4449	13.01	0.0537
Easy-H3		228.4	1.93×	**0.1651**	**0.7948**	**24.25**	**0.0249**
SARCache-H3	B	182.3	2.42×	0.3138	0.6236	18.12	0.0361
Sea-H3		183.5	2.40×	**0.3106**	**0.6264**	**18.23**	<u>0.0360</u>
Tea-style output hold		181.2	2.43×	<u>0.3122</u>	<u>0.6250</u>	<u>18.17</u>	**0.0359**
Taylor output predictor		180.2	2.45×	0.6886	0.3551	11.64	0.0742
Reduced steps (10)	adj.	239.3	1.84×	0.4899	0.5022	14.60	0.0395
SAR fixed $g4$	fixed	199.7	2.21×	0.2514	0.6966	20.37	0.0312
SAR fixed $g8$		199.5	2.21×	0.2885	0.6695	19.38	0.0323

Bold and underline mark the best and second-best displayed fidelity means within each test group. Ties share a rank. Speed is the ratio of mean reference and candidate times. Sea-H3 caches a packed-stack residual. Tea-style and Taylor rows use the shared SEA gate with output holding or output extrapolation. They are project adaptations. Easy-H3 and the reduced-step arm fall outside their calibration bands (Sec. B.2). Easy-H3 is displayed with A because it is inside the test A band of 215.46–238.14 s and has lower LPIPS than SARCache-A. Fixed arms use eight full calls, versus nine for SARCache-A. They are not equal-compute controls.